# Supplementary Material

When Simple Is Enough: Leakage-Free Benchmarking of Classical Machine Learning vs. Graph Neural Networks in Low-Data Bioactivity Prediction

This document holds the material that supports the manuscript without being part of its argument: the architecture chosen by the inner loop in every outer fold, and the detail of the screening and docking stage, whose results are hypotheses for experimental testing rather than conclusions of the benchmark. Every value is generated by the analysis pipeline and read from the same files as the manuscript.

## Benchmark detail

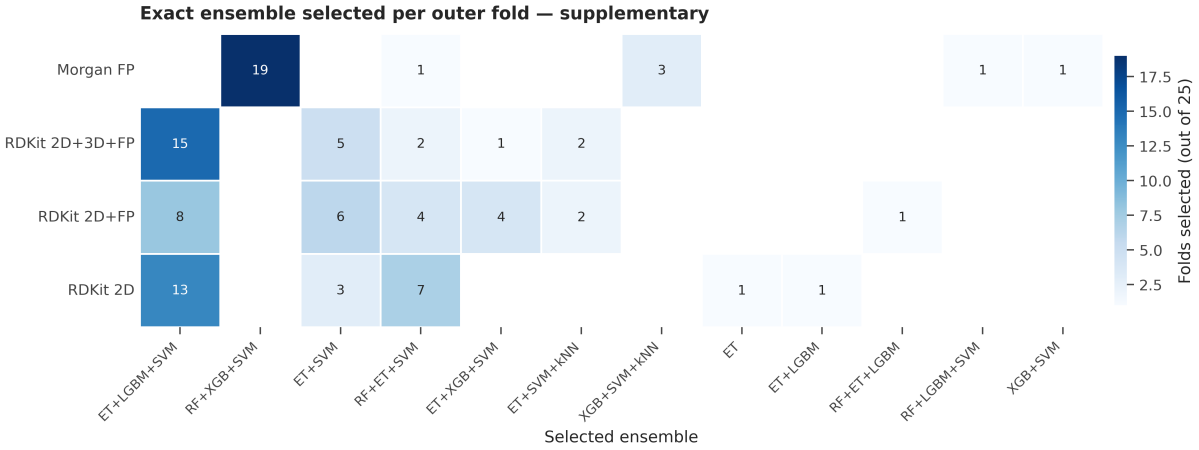



Figure S1: Stacking ensemble selected by the inner loop in each of the 25 outer folds, for each molecular representation. Cell values are the number of outer folds in which that exact combination of base learners (RF, random forest; ET, extra trees; LGBM, LightGBM; XGB, XGBoost; SVM, support vector regression; kNN, $k$-nearest neighbors) was chosen. No combination was chosen consistently, which shows that the inner search does not converge on a single architecture.

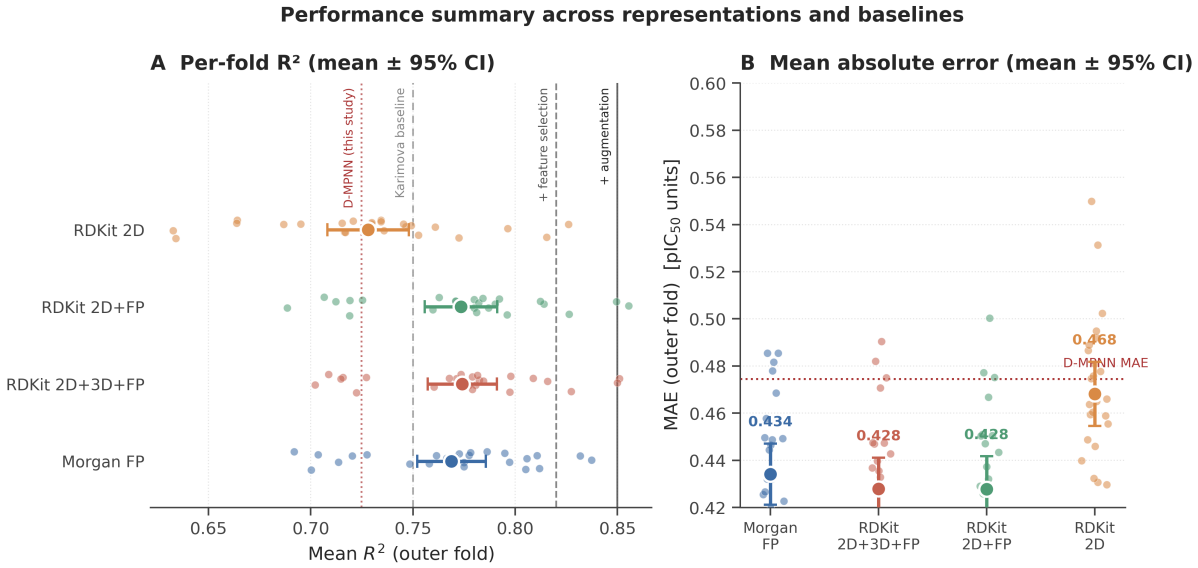



Figure S2: Per-fold performance of the four representations over the 25 outer folds. (A) $R^2$ of each fold (faint points), with the mean and its 95% confidence interval; (B) the same for the mean absolute error. The vertical lines mark the D-MPNN of this study and the three values reported by the original study, which come from a single hold-out split of a pipeline with the leaks quantified in Section 3.1 of the manuscript and are therefore shown for reference only, not as a paired comparison.

Table S1: Every pairwise comparison between the four molecular representations over the 25 shared outer folds, in both metrics. Differences are A minus B, the interval is the 90% confidence interval used by the equivalence test, $p$ is the Nadeau–Bengio corrected paired $t$-test, $p_{\text{Holm}}$ its step-down adjustment within the six comparisons of each metric, and $p_{\text{equiv}}$ the two one-sided tests against the $\pm 0.05$ margin.

| Metric | Comparison | Difference | 90% CI | $p$ | $p_{\text{Holm}}$ | $p_{\text{equiv}}$ |
|---|---|---|---|---|---|---|
| $R^2$ | Morgan vs RDKit 2D | +0.0408 | [+0.021, +0.061] | 0.0020 | 0.0081 | 0.2201 |
| $R^2$ | Morgan vs RDKit 2D+FP | −0.0047 | [−0.016, +0.007] | 0.5061 | 1.0000 | <0.0001 |
| $R^2$ | Morgan vs RDKit 2D+3D+FP | −0.0052 | [−0.016, +0.005] | 0.4006 | 1.0000 | <0.0001 |
| $R^2$ | RDKit 2D vs RDKit 2D+FP | −0.0454 | [−0.063, −0.028] | 0.0002 | 0.0009 | 0.3288 |
| $R^2$ | RDKit 2D vs RDKit 2D+3D+FP | −0.0460 | [−0.063, −0.029] | 0.0001 | 0.0008 | 0.3489 |
| $R^2$ | RDKit 2D+FP vs RDKit 2D+3D+FP | −0.0006 | [−0.005, +0.004] | 0.8243 | 1.0000 | <0.0001 |
| MAE | Morgan vs RDKit 2D | −0.0340 | [−0.051, −0.017] | 0.0025 | 0.0099 | 0.0625 |
| MAE | Morgan vs RDKit 2D+FP | +0.0065 | [−0.003, +0.016] | 0.2739 | 0.7689 | <0.0001 |
| MAE | Morgan vs RDKit 2D+3D+FP | +0.0063 | [−0.003, +0.016] | 0.2563 | 0.7689 | <0.0001 |
| MAE | RDKit 2D vs RDKit 2D+FP | +0.0405 | [+0.026, +0.054] | <0.0001 | 0.0002 | 0.1273 |
| MAE | RDKit 2D vs RDKit 2D+3D+FP | +0.0403 | [+0.028, +0.053] | <0.0001 | <0.0001 | 0.0980 |
| MAE | RDKit 2D+FP vs RDKit 2D+3D+FP | −0.0001 | [−0.005, +0.005] | 0.9664 | 0.9664 | <0.0001 |

Table S2: Composition of the stacking ensembles chosen by the inner loop, per representation. Each percentage is the share of the 25 outer folds in which that base learner entered the selected ensemble; stability is the share taken by the single most frequent exact combination. RF, random forest; ET, extra trees; LGBM, LightGBM; XGB, XGBoost; SVM, support vector regression; kNN, $k$-nearest neighbors.

| **Representation** | RF | ET | LGBM | XGB | SVM | kNN | **Most frequent** | **Stability** |
|---|---|---|---|---|---|---|---|---|
| Morgan FP | 84% | 4% | 4% | 92% | 100% | 12% | RF+XGB+SVM | 76.0% |
| RDKit 2D+3D+FP | 8% | 100% | 60% | 4% | 100% | 8% | ET+LGBM+SVM | 60.0% |
| RDKit 2D+FP | 20% | 100% | 36% | 16% | 96% | 8% | ET+LGBM+SVM | 32.0% |
| RDKit 2D | 28% | 100% | 56% | 0% | 92% | 0% | ET+LGBM+SVM | 52.0% |

Table S3: The four y-randomization tests of Rücker et al., each with 100 randomizations scored by 5-fold cross-validation. Scope "architecture" randomizes the stacking ensemble on all 2048 bits; scope "procedure" repeats the complete model-building procedure, including the out-of-fold permutation-importance selection that produces the deployed screening model, inside every randomization. The randomization is either a permutation of the response or a replacement of the descriptors by random ones. $Z$ is the distance from the real model to the mean of the randomized ones in units of their standard deviation; $p$ cannot fall below $1/(100+1)$.

| **Scope** | **Randomization** | $R^2$ (real) | $R^2$ (randomized) | $Z$ | $p$ |
|---|---|---|---|---|---|
| Architecture | Response permuted | 0.7787 | $-0.0136 \pm 0.0084$ | 93.9 | 0.0099 |
| Architecture | Random descriptors | 0.7787 | $-0.0136 \pm 0.0083$ | 95.7 | 0.0099 |
| Procedure | Response permuted | 0.7722 | $-0.0365 \pm 0.0169$ | 47.8 | 0.0099 |
| Procedure | Random descriptors | 0.7722 | $-0.0959 \pm 0.0322$ | 27.0 | 0.0099 |

Table S4: Each additional neural model against each classical representation over the 25 shared outer folds. Differences are the neural model minus the representation, so a negative value in $R^2$ means the neural model scored lower. $p_{\text{Holm}}$ is adjusted within the four representations of each model and metric.

| Metric | Comparison | Difference | 90% CI | $p_{\text{raw}}$ | $p_{\text{Holm}}$ | $p_{\text{equiv}}$ |
|---|---|---|---|---|---|---|
| $R^2$ | MLP on ECFP4 vs Morgan | −0.0598 | [−0.104, −0.015] | 0.0300 | 0.0965 | 0.6458 |
| $R^2$ | MLP on ECFP4 vs RDKit 2D | −0.0190 | [−0.069, +0.031] | 0.5193 | 0.5193 | 0.1492 |
| $R^2$ | MLP on ECFP4 vs RDKit 2D+FP | −0.0645 | [−0.110, −0.019] | 0.0241 | 0.0965 | 0.7029 |
| $R^2$ | MLP on ECFP4 vs RDKit 2D+3D+FP | −0.0651 | [−0.111, −0.019] | 0.0246 | 0.0965 | 0.7080 |
| $R^2$ | Ensemble of 5 D-MPNN vs Morgan | −0.0279 | [−0.051, −0.005] | 0.0505 | 0.1622 | 0.0582 |
| $R^2$ | Ensemble of 5 D-MPNN vs RDKit 2D | +0.0128 | [−0.016, +0.042] | 0.4598 | 0.4598 | 0.0199 |
| $R^2$ | Ensemble of 5 D-MPNN vs RDKit 2D+FP | −0.0326 | [−0.059, −0.006] | 0.0468 | 0.1622 | 0.1365 |
| $R^2$ | Ensemble of 5 D-MPNN vs RDKit 2D+3D+FP | −0.0332 | [−0.059, −0.007] | 0.0406 | 0.1622 | 0.1413 |
| $R^2$ | D-MPNN with RDKit 2D descriptors vs Morgan | −0.0681 | [−0.103, −0.033] | 0.0030 | 0.0121 | 0.8049 |
| $R^2$ | D-MPNN with RDKit 2D descriptors vs RDKit 2D | −0.0273 | [−0.068, +0.014] | 0.2657 | 0.2657 | 0.1764 |
| $R^2$ | D-MPNN with RDKit 2D descriptors vs RDKit 2D+FP | −0.0727 | [−0.112, −0.034] | 0.0040 | 0.0121 | 0.8353 |
| $R^2$ | D-MPNN with RDKit 2D descriptors vs RDKit 2D+3D+FP | −0.0733 | [−0.112, −0.035] | 0.0031 | 0.0121 | 0.8464 |
| $R^2$ | D-MPNN initialized from CheMeleon vs Morgan | −0.0074 | [−0.033, +0.018] | 0.6262 | 1.0000 | 0.0044 |
| $R^2$ | D-MPNN initialized from CheMeleon vs RDKit 2D | +0.0334 | [−0.002, +0.069] | 0.1216 | 0.4866 | 0.2164 |
| $R^2$ | D-MPNN initialized from CheMeleon vs RDKit 2D+FP | −0.0120 | [−0.040, +0.016] | 0.4753 | 1.0000 | 0.0155 |
| $R^2$ | D-MPNN initialized from CheMeleon vs RDKit 2D+3D+FP | −0.0126 | [−0.040, +0.015] | 0.4357 | 1.0000 | 0.0136 |
| MAE | MLP on ECFP4 vs Morgan | +0.0648 | [+0.032, +0.097] | 0.0024 | 0.0058 | 0.7770 |
| MAE | MLP on ECFP4 vs RDKit 2D | +0.0308 | [−0.006, +0.067] | 0.1649 | 0.1649 | 0.1896 |
| MAE | MLP on ECFP4 vs RDKit 2D+FP | +0.0712 | [+0.037, +0.106] | 0.0017 | 0.0058 | 0.8482 |
| MAE | MLP on ECFP4 vs RDKit 2D+3D+FP | +0.0711 | [+0.037, +0.105] | 0.0015 | 0.0058 | 0.8517 |
| MAE | Ensemble of 5 D-MPNN vs Morgan | +0.0258 | [+0.002, +0.050] | 0.0775 | 0.1550 | 0.0478 |
| MAE | Ensemble of 5 D-MPNN vs RDKit 2D | −0.0082 | [−0.030, +0.014] | 0.5262 | 0.5262 | 0.0017 |
| MAE | Ensemble of 5 D-MPNN vs RDKit 2D+FP | +0.0322 | [+0.008, +0.056] | 0.0305 | 0.0933 | 0.1083 |
| MAE | Ensemble of 5 D-MPNN vs RDKit 2D+3D+FP | +0.0321 | [+0.009, +0.055] | 0.0233 | 0.0933 | 0.0946 |
| MAE | D-MPNN with RDKit 2D descriptors vs Morgan | +0.0608 | [+0.028, +0.093] | 0.0037 | 0.0073 | 0.7136 |
| MAE | D-MPNN with RDKit 2D descriptors vs RDKit 2D | +0.0268 | [−0.002, +0.056] | 0.1250 | 0.1250 | 0.0906 |
| MAE | D-MPNN with RDKit 2D descriptors vs RDKit 2D+FP | +0.0672 | [+0.035, +0.100] | 0.0018 | 0.0053 | 0.8122 |
| MAE | D-MPNN with RDKit 2D descriptors vs RDKit 2D+3D+FP | +0.0671 | [+0.036, +0.098] | 0.0012 | 0.0048 | 0.8209 |
| MAE | D-MPNN initialized from CheMeleon vs Morgan | +0.0075 | [−0.020, +0.035] | 0.6431 | 1.0000 | 0.0067 |
| MAE | D-MPNN initialized from CheMeleon vs RDKit 2D | −0.0265 | [−0.055, +0.002] | 0.1291 | 0.5164 | 0.0886 |
| MAE | D-MPNN initialized from CheMeleon vs RDKit 2D+FP | +0.0139 | [−0.014, +0.042] | 0.3964 | 1.0000 | 0.0174 |
| MAE | D-MPNN initialized from CheMeleon vs RDKit 2D+3D+FP | +0.0138 | [−0.012, +0.040] | 0.3729 | 1.0000 | 0.0128 |

Table S5: Comparisons under scaffold-grouped folds: every neural model against every classical representation, and the classical representations against each other. The unadjusted $p$ is given beside the adjusted one because the 90% intervals quoted in the manuscript are not corrected for multiplicity, which is why an interval can exclude zero while the adjusted $p$ is far from significance.

| **Metric** | **Comparison** | **Difference** | **90% CI** | $p_{\text{raw}}$ | $p_{\text{Holm}}$ | $p_{\text{equiv}}$ |
|---|---|---|---|---|---|---|
| $R^2$ | D-MPNN, default vs Morgan | −0.0729 | [−0.144, −0.002] | 0.0919 | 0.3676 | 0.7070 |
| $R^2$ | D-MPNN, default vs RDKit 2D | +0.0291 | [−0.045, +0.103] | 0.5067 | 0.7607 | 0.3161 |
| $R^2$ | D-MPNN, default vs RDKit 2D+FP | −0.0495 | [−0.125, +0.026] | 0.2737 | 0.7607 | 0.4954 |
| $R^2$ | D-MPNN, default vs RDKit 2D+3D+FP | −0.0528 | [−0.130, +0.024] | 0.2536 | 0.7607 | 0.5243 |
| $R^2$ | DNN baseline vs Morgan | −0.1053 | [−0.215, +0.004] | 0.1118 | 0.4470 | 0.8028 |
| $R^2$ | DNN baseline vs RDKit 2D | −0.0033 | [−0.114, +0.107] | 0.9594 | 0.9594 | 0.2380 |
| $R^2$ | DNN baseline vs RDKit 2D+FP | −0.0819 | [−0.200, +0.036] | 0.2453 | 0.6965 | 0.6765 |
| $R^2$ | DNN baseline vs RDKit 2D+3D+FP | −0.0852 | [−0.204, +0.034] | 0.2322 | 0.6965 | 0.6914 |
| $R^2$ | DNN with selection and augmentation vs Morgan | −0.0837 | [−0.159, −0.008] | 0.0703 | 0.2812 | 0.7735 |
| $R^2$ | DNN with selection and augmentation vs RDKit 2D | +0.0183 | [−0.072, +0.109] | 0.7328 | 0.7328 | 0.2777 |
| $R^2$ | DNN with selection and augmentation vs RDKit 2D+FP | −0.0603 | [−0.147, +0.026] | 0.2445 | 0.6514 | 0.5796 |
| $R^2$ | DNN with selection and augmentation vs RDKit 2D+3D+FP | −0.0636 | [−0.149, +0.022] | 0.2171 | 0.6514 | 0.6054 |
| $R^2$ | Ensemble of D-MPNN vs Morgan | −0.0392 | [−0.106, +0.028] | 0.3259 | 0.9778 | 0.3919 |
| $R^2$ | Ensemble of D-MPNN vs RDKit 2D | +0.0629 | [−0.013, +0.139] | 0.1684 | 0.6736 | 0.6131 |
| $R^2$ | Ensemble of D-MPNN vs RDKit 2D+FP | −0.0157 | [−0.092, +0.060] | 0.7272 | 1.0000 | 0.2242 |
| $R^2$ | Ensemble of D-MPNN vs RDKit 2D+3D+FP | −0.0190 | [−0.096, +0.058] | 0.6773 | 1.0000 | 0.2493 |
| $R^2$ | D-MPNN initialized from CheMeleon vs Morgan | +0.0065 | [−0.038, +0.051] | 0.8035 | 0.9247 | 0.0538 |
| $R^2$ | D-MPNN initialized from CheMeleon vs RDKit 2D | +0.1086 | [+0.051, +0.166] | 0.0035 | 0.0142 | 0.9531 |
| $R^2$ | D-MPNN initialized from CheMeleon vs RDKit 2D+FP | +0.0300 | [−0.019, +0.079] | 0.3082 | 0.9247 | 0.2471 |
| $R^2$ | D-MPNN initialized from CheMeleon vs RDKit 2D+3D+FP | +0.0267 | [−0.025, +0.078] | 0.3851 | 0.9247 | 0.2238 |
| MAE | D-MPNN, default vs Morgan | +0.0486 | [−0.002, +0.100] | 0.1160 | 0.4638 | 0.4820 |
| MAE | D-MPNN, default vs RDKit 2D | −0.0260 | [−0.081, +0.029] | 0.4298 | 0.8935 | 0.2332 |
| MAE | D-MPNN, default vs RDKit 2D+FP | +0.0335 | [−0.023, +0.090] | 0.3176 | 0.8935 | 0.3096 |
| MAE | D-MPNN, default vs RDKit 2D+3D+FP | +0.0348 | [−0.021, +0.091] | 0.2978 | 0.8935 | 0.3228 |
| MAE | DNN baseline vs Morgan | +0.0159 | [−0.030, +0.061] | 0.5565 | 1.0000 | 0.1058 |
| MAE | DNN baseline vs RDKit 2D | −0.0588 | [−0.115, −0.002] | 0.0885 | 0.3541 | 0.6037 |
| MAE | DNN baseline vs RDKit 2D+FP | +0.0007 | [−0.056, +0.058] | 0.9835 | 1.0000 | 0.0763 |
| MAE | DNN baseline vs RDKit 2D+3D+FP | +0.0020 | [−0.054, +0.058] | 0.9523 | 1.0000 | 0.0790 |
| MAE | DNN with selection and augmentation vs Morgan | +0.0125 | [−0.050, +0.075] | 0.7378 | 1.0000 | 0.1587 |
| MAE | DNN with selection and augmentation vs RDKit 2D | −0.0622 | [−0.132, +0.007] | 0.1383 | 0.5532 | 0.6170 |
| MAE | DNN with selection and augmentation vs RDKit 2D+FP | −0.0027 | [−0.074, +0.068] | 0.9485 | 1.0000 | 0.1328 |
| MAE | DNN with selection and augmentation vs RDKit 2D+3D+FP | −0.0014 | [−0.072, +0.069] | 0.9729 | 1.0000 | 0.1250 |
| MAE | Ensemble of D-MPNN vs Morgan | +0.0244 | [−0.020, +0.069] | 0.3561 | 1.0000 | 0.1659 |
| MAE | Ensemble of D-MPNN vs RDKit 2D | −0.0503 | [−0.106, +0.005] | 0.1357 | 0.5427 | 0.5037 |
| MAE | Ensemble of D-MPNN vs RDKit 2D+FP | +0.0092 | [−0.048, +0.067] | 0.7866 | 1.0000 | 0.1181 |
| MAE | Ensemble of D-MPNN vs RDKit 2D+3D+FP | +0.0105 | [−0.047, +0.068] | 0.7584 | 1.0000 | 0.1264 |
| MAE | D-MPNN initialized from CheMeleon vs Morgan | +0.0005 | [−0.030, +0.032] | 0.9775 | 1.0000 | 0.0058 |
| MAE | D-MPNN initialized from CheMeleon vs RDKit 2D | −0.0741 | [−0.109, −0.039] | 0.0014 | 0.0056 | 0.8742 |
| MAE | D-MPNN initialized from CheMeleon vs RDKit 2D+FP | −0.0146 | [−0.052, +0.022] | 0.5033 | 1.0000 | 0.0570 |
| MAE | D-MPNN initialized from CheMeleon vs RDKit 2D+3D+FP | −0.0133 | [−0.052, +0.026] | 0.5632 | 1.0000 | 0.0603 |
| $R^2$ | Morgan vs RDKit 2D | +0.1020 | — | 0.0002 | 0.0015 | — |
| $R^2$ | Morgan vs RDKit 2D+FP | +0.0235 | — | 0.3332 | 0.9997 | — |
| $R^2$ | Morgan vs RDKit 2D+3D+FP | +0.0202 | — | 0.4151 | 0.9997 | — |
| $R^2$ | RDKit 2D vs RDKit 2D+FP | −0.0786 | — | 0.0028 | 0.0139 | — |
| $R^2$ | RDKit 2D vs RDKit 2D+3D+FP | −0.0819 | — | 0.0031 | 0.0139 | — |
| $R^2$ | RDKit 2D+FP vs RDKit 2D+3D+FP | −0.0033 | — | 0.6539 | 0.9997 | — |

## Virtual screening and molecular docking

### Ligand efficiency metrics

Four efficiency metrics contextualize a predicted potency against the physicochemical load that carries it, which mitigates the risk of prioritizing compounds of high predicted potency and unfavourable properties for development [Hopkins et al., 2004]. Ligand efficiency (LE) is the binding energy per heavy atom count (HAC) [Hopkins et al., 2014]:

$$\mathrm{LE} = \frac{1.37 \times pIC_{50}}{\mathrm{HAC}} \tag{1}$$

The factor of 1.37 converts a dissociation constant into a binding free energy, so substituting $pIC_{50}$ for $pK_d$ assumes the two are interchangeable, which holds only under the Cheng–Prusoff conditions [Cheng and Prusoff, 1973]. The assumption is inherited from the data set, whose potencies are $IC_{50}$ values measured in assays whose substrate concentrations are not reported, and it is one reason none of these metrics was used as a filter. The binding efficiency index (BEI) substitutes the heavy-atom count with the molecular weight (MW) [Abad-Zapatero and Metz, 2005]:

$$\mathrm{BEI} = \frac{pIC_{50}}{\mathrm{MW}} \times 1000 \tag{2}$$

Lipophilic ligand efficiency balances potency against lipophilicity, to mitigate the risk of non-specific binding, as the difference between predicted potency and the calculated octanol–water partition coefficient ($\mathrm{LLE} = pIC_{50} - \log P$, Wildman–Crippen method in RDKit). The surface efficiency index evaluates potency relative to the topological polar surface area (TPSA), as a proxy for membrane permeability ($\mathrm{SEI} = (pIC_{50}/\mathrm{TPSA}) \times 100$). Because composite indices combining logarithmic potency values with linear physical parameters can introduce size-dependent mathematical artifacts [Shultz, 2013], the four metrics were computed from the predicted $pIC_{50}$ and reported alongside it for interpretation; candidates were ranked and selected by predicted potency alone.

### Docking protocol

The structure-based check complements the ligand-based predictions by assessing complementarity with the binding site, which mitigates the risk of false positives produced by scaffold memorization. AutoDock Vina v1.2.7 [Das et al., 2026, Trott and Olson, 2010] employs an Iterated Local Search optimizer and a hybrid scoring function, calibrated against PDBbind data, to guide flexible ligands into optimal binding conformations [Eberhardt et al., 2021].

Three-dimensional receptor coordinates were retrieved from the Protein Data Bank (*Toxoplasma gondii* DHFR, PDB ID: 6AOG) [wwPDB consortium, 2019, Berman et al., 2003]. The receptor was prepared as a rigid apo structure: all non-protein records (crystallographic waters, ions, cofactors including the bound NADPH, and the co-crystallized ligand) were removed, retaining both protein chains. Hydrogens were added at pH 7.4 and the structure was converted to the PDBQT format with OpenBabel in rigid mode [Babel, 2011]; keeping the polar hydrogens types the backbone N–H groups as hydrogen-bond donors, which the Vina scoring function distinguishes from acceptors (the partial charges written to the file are not used by this scoring function). While excluding the NADPH cofactor omits a potential source of active-site polarization, this apo-receptor approximation is a common simplification in structure-based virtual screening. The Vina search space was centered on the coordinates of the native CP6 ligand in chain B ($x = 3.689, y = 39.992, z = -62.818$) with a cubic grid box of $20.0 \times 20.0 \times 20.0$ Å and an exhaustiveness parameter of 32, which permitted unrestricted exploration of ligand orientations and rotatable bonds within the site [Trott and Olson, 2010].

Prior to screening the machine learning candidates, the docking protocol was checked by re-docking the co-crystallized CP6 ligand into the TgDHFR active site. Because the PDB file of

the ligand has no bond orders, they were assigned from the known structure of pyrimethamine before the docking input was built, and the starting conformer was generated independently of the crystallographic pose. The top-ranked pose reproduced the crystallographic pose with a symmetry-aware heavy-atom RMSD of 1.11 Å (centroid distance 0.35 Å), below the 2.0 Å criterion adopted for validation. The search box was centered on the crystallographic ligand, so the check confirms that the grid parameters and the docking setup are consistent with the known binding site, but it does not demonstrate blind pose prediction, and it was run on a single ligand.

Following validation, Vina evaluated all 17 candidates of the screening step (the 10 top-ranked compounds plus the reference entries). For each compound the largest fragment was retained to remove counter-ions, hydrogens were added, a single 3D conformer was generated with ETKDGv3 (fixed seed) and minimized with MMFF, and a flexible-ligand PDBQT file was prepared with Meeko, keeping the protonation state of the input structure. Docking used a fixed random seed and returned the 3 best poses. Poses were ranked by the predicted binding free energy ($\Delta G$) in kcal/mol, where lower energies denote stronger predicted binding [Feng et al., 2023]; this is an empirical, rigid-receptor estimate and not a measured thermodynamic quantity. The docking ligand efficiency was computed as the best score per heavy atom. To systematically evaluate the consensus between the ligand-based and structure-based approaches, we established quantitative thresholds. Compounds were classified as validated consensus hits only if they exhibited both high ML-predicted potency ($pIC_{50} > 6.5$) and strong docking affinity ($\Delta G < -8.0$ kcal/mol). Compounds meeting the ML threshold but failing the docking criterion were flagged as potential ML false positives. Conversely, compounds with $pIC_{50} < 6.0$ were labeled validated negatives if $\Delta G > -7.5$ kcal/mol and potential ML false negatives if $\Delta G < -8.0$ kcal/mol; all remaining compounds were classed as neutral. Finally, to quantify the monotonic agreement between the two orthogonal methods without assuming linearity, a Spearman rank correlation analysis was conducted between the continuous $pIC_{50}$ predictions and the Vina docking scores (reported in the manuscript).

## Screening results

Table S6: Top 10 FDA-approved candidates predicted by the classical ML model, ranked by predicted $pIC_{50}$ and evaluated across standard ligand efficiency metrics. LE, ligand efficiency; BEI, binding efficiency index; LLE, lipophilic ligand efficiency; SEI, surface efficiency index.

| **Rank** | **Compound Name** | **Predicted $pIC_{50}$** | **LE** | **BEI** | **LLE** | **SEI** |
|---|---|---|---|---|---|---|
| 1 | Trimetrexate Glucuronate | 7.9272 | 0.40 | 21.46 | 5.19 | 6.74 |
| 2 | Methotrexate | 7.4448 | 0.31 | 16.38 | 7.18 | 3.54 |
| 3 | Triamterene | 7.0081 | 0.51 | 27.67 | 6.17 | 5.41 |
| 4 | Pralatrexate | 7.0034 | 0.27 | 14.67 | 6.02 | 3.38 |
| 5 | Pyrimethamine | 6.5705 | 0.53 | 26.42 | 4.05 | 8.44 |
| 6 | Folic Acid | 6.3379 | 0.27 | 14.36 | 6.38 | 2.97 |
| 7 | Naldemedine Tosylate | 6.0892 | 0.20 | 10.67 | 2.61 | 4.31 |
| 8 | Bendamustine Hydrochloride | 6.0850 | 0.36 | 16.98 | 2.82 | 10.43 |
| 9 | Levomefolate Calcium | 6.0732 | 0.25 | 13.22 | 6.33 | 3.00 |
| 10 | Letermovir | 5.9681 | 0.20 | 10.42 | 0.26 | 7.67 |

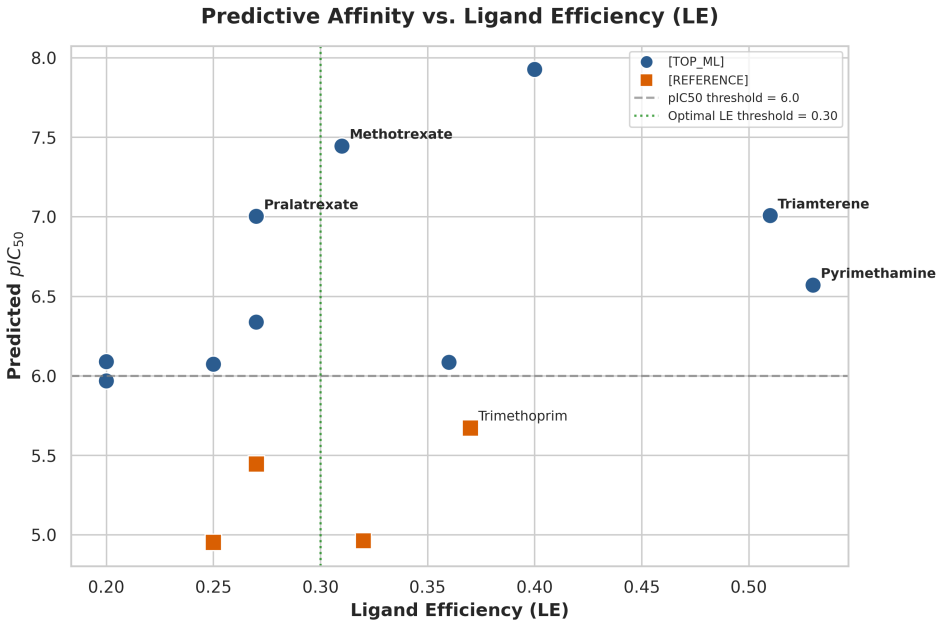



Figure S3: Predicted $pIC_{50}$ versus ligand efficiency for the candidates advanced to docking. Blue: the ten top-ranked compounds by predicted potency; orange: reference compounds from the original study. The dashed line marks $pIC_{50} = 6.0$ and the dotted line LE = 0.30.

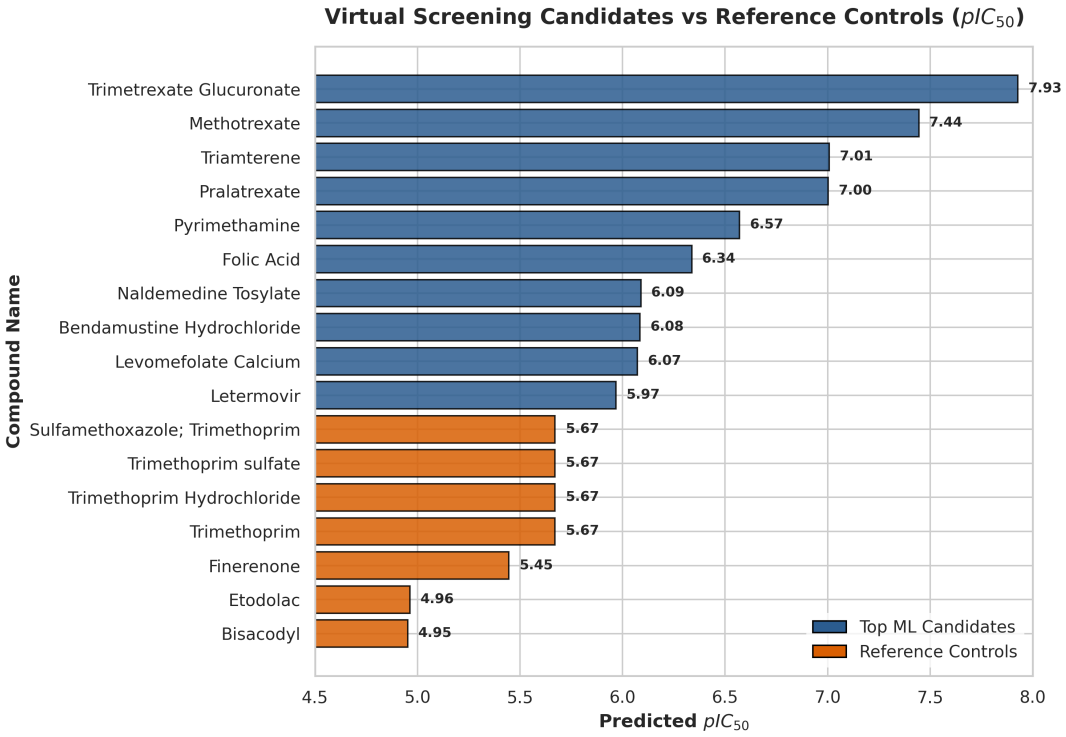



Figure S4: Predicted $pIC_{50}$ of the screening candidates next to the reference controls of the original study. Compounds that belong to the training partition are fitted values and not predictions on unseen molecules.

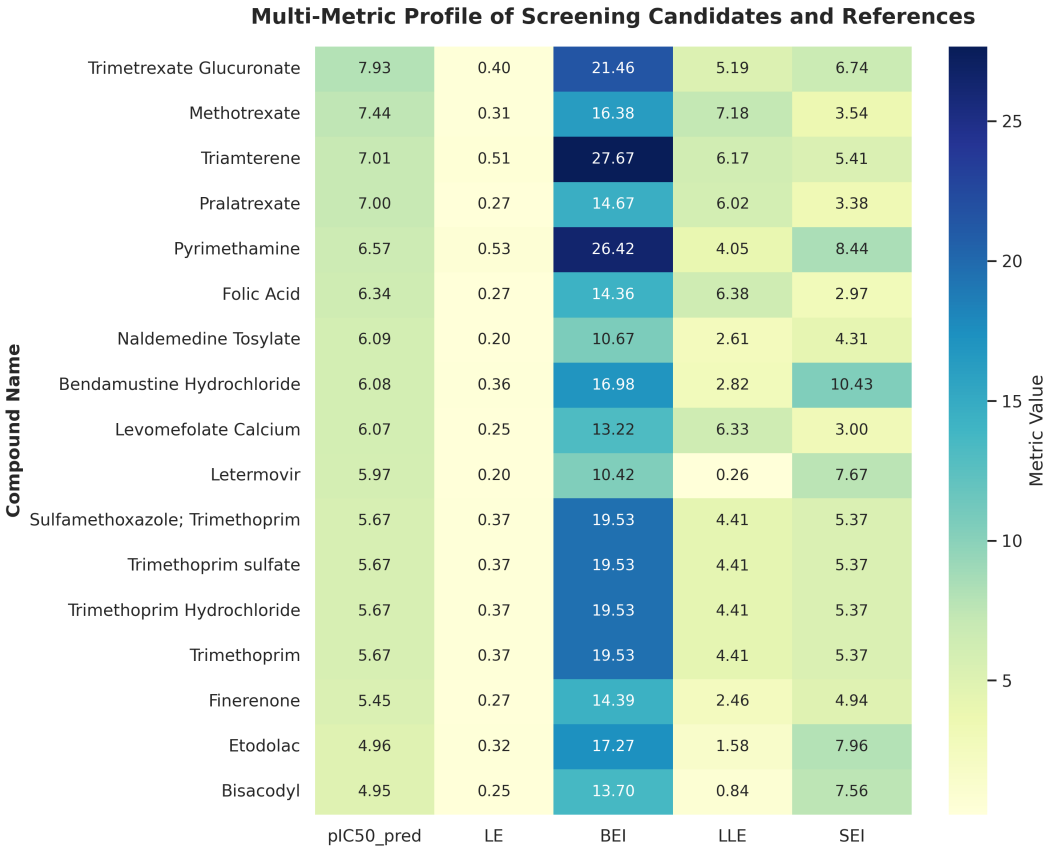



Figure S5: Efficiency metrics of the screening candidates and the reference controls. Each metric is scaled across compounds, so the colour shows a compound's standing within this set and not an absolute quality.

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